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Indian Institute of Technology Jodhpur

MTech in Thermo-Fluids Engineering
Department of Mechanical Engineering

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M. Tech. in Thermo-Fluids Engineering (TFE)
Department of Mechanical Engineering

Background:

Thermo-Fluids forms the scientific foundation of a wide range of engineering systems involving fluid transport, thermal management, energy conversion, combustion, and environmental processes. Rapid advancements in sectors such as renewable energy, green refrigeration & air conditioning technologies, automotive engineering, water technologies, chemical processing, and advanced manufacturing are strongly reliant on engagement of engineers with expertise in fluid dynamics, heat transfer, thermodynamics, and combustion. Modern engineering challenges require the understanding and control of complex flow phenomena, efficient thermal systems, multiphase flows, and sustainable energy technologies through advanced computational and experimental approaches.

The MTech program in Thermo-Fluids at IIT Jodhpur is designed to address these needs by providing advanced training in the fundamental and applied aspects of thermal and fluid sciences. The program covers areas such as fluid dynamics, heat transfer, combustion, water resource technologies, natural refrigerant technologies, non-conventional cooling technologies, and computational modelling, while fostering interdisciplinary interactions with emerging fields including energy systems, materials, manufacturing, and medical technologies. Through a combination of core courses, electives, and research projects, students will develop the analytical, computational, and problem-solving skills required to address complex engineering challenges.

Furthermore, national priorities related to energy security, sustainability, water resources, health and technological self-reliance have created a growing need for highly trained professionals in thermo-fluid sciences. The program will contribute to these goals by preparing graduates for advanced research, innovation, and leadership roles in academia, government laboratories, and a broad spectrum of industries. By integrating fundamental science with modern engineering tools, the program supports India's long-term objectives in sustainable technologies and innovation-driven growth.

Objectives of the Program:

- To develop professionals through analytical and computational skills that enables individual to handle research and industrial problems in the domain of thermo-fluids engineering.
- To impart advanced knowledge of fluid dynamics, thermodynamics, heat transfer, combustion, HVAC, and multiphase flows for the analysis, design, and optimization of engineering systems.
- To train students in modern computational techniques, including computational fluid dynamics (CFD), numerical modelling, simulation, smart energy systems, and data-driven approaches for solving thermo-fluid engineering problems.
- To develop competence in the design and performance evaluation of thermal and fluid systems such as energy conversion devices, thermal management systems, fluid transport networks, renewable energy systems, green refrigeration & air conditioning technologies, heat pumps, natural refrigerants, non-conventional cooling technologies and process equipment.

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- To enable students to integrate theoretical principles with experimental methods, diagnostics, instrumentation, and computational tools for comprehensive analysis and validation of thermo-fluid systems.
- To foster interdisciplinary thinking by applying thermo-fluid principles to emerging areas such as sustainable energy, water technologies, advanced manufacturing, healthcare and environmental engineering.
- To inculcate strong research aptitude, critical thinking, and problem-solving abilities, preparing students for doctoral studies, industrial R&D, and innovation-driven careers.
- To equip graduates with the skills required by academia, government laboratories, and industries including chemical processing, manufacturing, and allied sectors.

Graduate Attributes:

- Ability to undertake academic and industrial research in the areas of Thermal and Fluids Engineering like automobiles, thermal power, nuclear energy, gas turbines, solar energy, space science, defence, environmental engineering etc.
- Ability to solve problems in design of thermal and fluid equipment/systems using analytical, experimental and computational techniques
- Equipped to contribute to the multi-physics and interdisciplinary research interfacing with thermal and fluid systems
- Ability to express ideas in the written and oral formats
- Development of appreciation and commitment to professional ethics

Learning Outcomes:

At the end of program, a student is expected to have:

- Strong fundamentals in incompressible/compressible fluid flow, thermodynamics and heat transfer
- Analytical, computational and experimental skills to solve problems in multi-scale heat and fluid flow
- Skills to design, develop and integrate systems/equipment involving heat and fluid flow
- Competence to identify and analyze interdisciplinary systems/devices involving thermos-fluids
- Expertise in one of the application areas e.g. solar thermal, solar cooling, combustion, heat exchanger design, green refrigeration & air conditioning, heat pumps, aerodynamics, thermal management of electronic systems, high speed flows etc.
- Skills to communicate technical and scientific ideas in the domain of Thermal and Fluids Engineering

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Eligible Branches and Justification:

Regular students: BTech, B.E. (Mechanical Engineering, Production Engineering, Industrial Engineering, Production and Industrial Engineering, Aerospace Engineering, Mechatronics Engineering, Applied Mechanics, Automobile Engineering, Aeronautical Engineering, Power plant engineering, Nanotechnology, Materials Engineering, Instrumentation Engineering)

Sponsored students: BTech, B.E. (Mechanical Engineering, Production Engineering, Industrial Engineering, Production and Industrial Engineering, Aerospace Engineering, Mechatronics Engineering, Applied Mechanics, Automobile Engineering, Aeronautical Engineering, Power plant engineering, Nanotechnology, Materials Engineering, Instrumentation Engineering)

Student Intake Number and Strategy:

A maximum of 16 students is planned to be admitted to this program. These include 11 GATE-qualified students (MoE fellowship) and 5 sponsored students.

Overall Program Philosophy:

The MTech program in Thermo-Fluids Engineering at IIT Jodhpur is designed to develop strong foundations in fluid mechanics, thermodynamics, heat and mass transfer, and computational methods, with an emphasis on physical understanding, quantitative analysis, and engineering applications. The program prepares students to analyze, model, design, and optimize thermo-fluid systems encountered in energy, water, transportation, healthcare, manufacturing, and process industries, while addressing contemporary challenges related to efficiency, sustainability, and technological innovation.

This is enabled through: (1) a core curriculum covering engineering mathematics, fluid flow, heat and mass transfer, experimental methods, computational techniques, and laboratory-based learning to provide breadth and essential hands-on skills, and, (2) a set of electives offering depth and flexibility in advanced areas such as thermodynamics, turbulent and compressible flows, combustion, computational fluid dynamics, heat exchanger design, nanoscale thermal systems, and probabilistic engineering methods. The program further promotes interdisciplinary learning by integrating theoretical foundations with modern computational and experimental tools for solving complex thermo-fluid engineering problems.

Objectives of the Core Program:

The core program is designed to provide students with a strong foundation in the fundamental and advanced principles of thermo-fluid sciences, including fluid mechanics, thermodynamics, heat and mass transfer, and modern experimental and computational techniques. It aims to develop the analytical, numerical, and practical skills required for the modelling, analysis, and design of thermo-fluid systems encountered in diverse engineering applications. It is to be noted that a total of 13 credits is to be completed through the program core route. This is achieved through a set of core theory courses and laboratory-based training, as listed below.

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Program Core (Compulsory) Courses:

S. No.	Course Number	Course Title	L-T-P	Credits
1	MEL7XX0	Engineering Mathematics	3-0-0	3
2	MEL7XX0	Incompressible Fluid Flow	3-0-0	3
3	MEL7XX0	Applied Heat and Mass Transfer	3-0-0	3
4	MEL7XX0	Experimental and Computational Techniques	3-0-0	3
5	MEL7XX0	Thermo-Fluids Engineering Lab	0-0-2	1

Objectives of the Elective Program:

The elective structure of the MTech in Thermo-Fluids Engineering is designed to provide students with advanced knowledge and specialization opportunities in key areas of thermal and fluid sciences. The elective framework offers both depth and flexibility, allowing students to tailor their academic pathway in accordance with their research interests and career aspirations. It is to be noted that a total of 12 credits is to be completed through the program elective route. The list of program electives applicable to the MTech in Thermo-Fluids Engineering is provided below.

Electives Courses:

S. No.	Course Number	Course Title	L-T-P	Credits
1	MEL7XX0	Engineering Thermodynamics	3-0-0	3
2	MEL7XX0	Convective Heat and Mass Transfer	3-0-0	3
3	MEL7XX0	Conduction and Radiation	3-0-0	3
4	MEL7XX0	Turbulent Flows	3-0-0	3
5	MEL7XX0	Computational Fluid Dynamics and Heat Transfer	3-0-0	3
6	MEL7XX0	Compressible Fluid Flows	3-0-0	3
7	MEL7XX0	Nanosensors	3-0-0	3
8	MEL7XX0	Probabilistic Methods in Engineering Design*	3-0-0	3
9	MEL7XX0	Combustion Technology	3-0-0	3
10	MEL7XX0	Design of Heat Exchange Equipment	3-0-0	3
11	MEL7XX0	Thermal Environmental Engineering	3-0-0	3
12	MEL7XX0	Machine Learning for Thermofluids	3-0-0	3
13	MEL7XX0	Special Topics	3-0-0	3

Note – Total number of program electives can be minimum 8 & maximum 12 (based on 4 slots in the curriculum earmarked for electives); for accommodating courses from new faculty, the total number of program electives can be extended up to 16

*to be offered as Open Elective

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Program's Structure:

S. No.	Course Type	Credits
1	Core	13
2	Program Electives	12
3	M. Tech. Thesis I	16
4	M. Tech. Thesis II	16
5	M. Tech. Thesis summer	8
Graded		**65**
6	Non-Graded Courses (2 credits)	0
Total		**65**

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Course Structure: Semester-wise Distribution of Credits
for MTech in Thermo-Fluids Engineering students for the batches from Academic Year 2026-27

Cat.	*Course Number, Course Title*		*L-T-P*	*Credits*	*Cat.*	*Course Number, Course Title*		*L-T-P*	*Credits*
I Semester					**II Semester**				
C	MEL7XX0	Engineering Mathematics	3-0-0	3	C	MEL7XX0	Experimental and Computational Techniques	3-0-0	3
C	MEL7XX0	Incompressible Fluid Flow	3-0-0	3	C	MEL7XX0	Thermo-Fluids Engineering Lab	0-0-2	1
C	MEL7XX0	Applied Heat and Mass Transfer	3-0-0	3	E	MEL7XX0	Elective	3-0-0	3
E	MEL7XX0	Elective	3-0-0	3	E	MEL7XX0	Elective	3-0-0	3
NG1	LAN7XX0	Technical Communication	1-0-0	S/X	E	MEL7XX0	Elective	3-0-0	3
					NG2	OAL7XX0	Innovation and IP Management	1-0-0	S/X
Total				**12**	*Total*				**13**
					Summer Term				
					T	MES7XX0	Thesis		8
					Total				**8**
III Semester					IV Semester				
T	MET7XX0	Thesis		**16**	T	MET7XX0	Thesis		**16**

S.No.	*Category*	*Course Category Title*	*Total Courses*	*Total Credits*
1	C	*COMPULSORY*	5	13
2	E	*ELECTIVES*	4	12
3	NG	*Non-Graded*	2	0
3	T	Thesis	**32+8**	32+8
		Total		**65**

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***for Direct students and students with different background**

Program compulsory (C)	Electives (E)
Engineering Mathematics Incompressible Fluid Flow Applied Heat and Mass Transfer Experimental and Computational Techniques Thermo-Fluids Engineering Lab	Engineering Thermodynamics Convective Heat and Mass Transfer Conduction and Radiation Turbulent Flows Computational Fluid Dynamics and Heat Transfer Compressible Fluid Flows Nanosensors Probabilistic Methods in Engineering Design* Combustion Technology Design of Heat Exchange Equipment Thermal Environmental Engineering Machine Learning for Thermofluids Special Topics *to be offered as Open Elective

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Department of Mechanical Engineering

MTech in Thermo-Fluids Engineering Courses

1. The following five courses are compulsory for MTech in Thermo-Fluids Engineering students of the department:

    1. **Engineering Mathematics (Existing course)**
    2. **Incompressible Fluid Flow (Existing course)**
    3. **Applied Heat and Mass Transfer (Existing course)**
    4. **Experimental and Computational Techniques (New Course)[1]**
    5. **Thermo-Fluids Engineering Lab (New Course)[§]**

2. MTech in Thermo-Fluids Engineering students shall have the option to choose four courses from the following list of courses:

    1. **Engineering Thermodynamics (Existing Course)**
    2. **Convective Heat and Mass Transfer (Existing Course)**
    3. **Conduction and Radiation (Existing course)**
    4. **Turbulent Flows (Existing Course)**
    5. **Computational Fluid Dynamics and Heat Transfer (Existing Course)**
    6. **Compressible Fluid Flows (Existing Course)**
    7. **Nanosensors (Existing Course)**
    8. **Probabilistic Methods in Engineering Design* (Existing Course)**
    9. **Combustion Technology (Existing Course)**
    10. **Design of Heat Exchange Equipment (Existing Course)**
    11. **Thermal Environmental Engineering (New Course)**
    12. **Machine Learning for Thermofluids (Existing Course)**
    13. **Special Topics (New Course)**

    *to be offered as Open Elective

The syllabus and checklists for the existing as well as new courses are given in the subsequent pages.

[1]Separate theory and lab courses introduced in lieu of formerly combined course on Experimental and Computational Techniques

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Title	Engineering Mathematics	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	MTech and PhD	Type	C
Prerequisite			

Objectives:

1. Equip students with a comprehensive understanding of advanced mathematical tools (vector calculus, linear algebra, differential equations) and their application in modeling and solving complex physical and engineering systems using ordinary and partial differential equations.
2. Introduce foundational concepts in probability and statistics to enable students to interpret data and analyze uncertainty in engineering processes.

Learning Outcomes:

1. Students will proficiently apply vector calculus, tensor analysis, matrix operations, and both analytical and numerical methods for solving ODEs and PDEs to address engineering problems involving multivariable systems, transformations, and physical contexts like vibrations and heat transfer.
2. Students will effectively use statistical methods to analyze experimental data, conduct hypothesis testing, and develop simple predictive models for engineering applications.

Contents:

1. Vectors & Tensors: Vector Algebra; Functions of several variables; Curves and surfaces; Multiple integrals; Gauss (divergence), Stokes and Green's theorems; Tensor notation, transformation and operations **[9 lectures]**
2. Linear Algebra: Matrix operations; LU factorization; Reduced row echelon form and rank; Independence, basis & dimension; Orthogonality, projection and least square approximation; Basis Vectors; Determinant properties; condition number; Eigenvalues and Eigenvectors - including examples; Singular Value decomposition **[8 lectures]**
3. ODEs: First order ODEs; second order linear ODEs; system of ODEs; phase-plane relationship; series solutions of ODEs; Laplace transforms; Fourier series; Sturm-Liouville problems; Fourier transforms; Examples - mass-spring-damper system, multi-degree of freedom discrete vibration systems **[8 Lectures]**
4. Numerical ODE: Initial value problem using Euler's Method, Runge-Kutta Method, Integrating Higher order ODE using system of ODEs, stiff problems, Boundary value problem using shooting method **[4 Lectures]**
5. PDEs: Characterization of PDEs, Separation of Variables, Examples - Unsteady heat conduction equation, string vibrations - Eigen modes etc, Cauchy equilibrium equation-applied on thin elastic plate **[4 Lectures]**
6. Probability and Statistics: Experiments, outcomes, events, central limit theorem, random variables, probability distributions, distributions of several random variables, random sampling, point estimation, confidence intervals, Testing hypotheses, Goodness of fit, regression, correlations. Examples from thermofluids - temperature data in Jodhpur, rainfall data **[6 Lectures]**

Textbook:

1. Kreyszig, E. (2011). Advanced engineering mathematics (10th ed.). John Wiley & Sons.

Reference books:

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1. Abeyratne, R. (2021). Volume 1. A brief review of some mathematical preliminaries (Version 1.0: 2 Dec 2006; updated 09 Jul 2021).
2. Strang, Gilbert (2016). Introduction to Linear Algebra. 5th ed., Wellesley-Cambridge Press.
3. Borisenko, A. I., and I. E. Tarapov (1979). Vector and Tensor Analysis with Applications. Translated by Richard A. Silverman, Dover Publications.
4. Chapra, S. C., & Canale, R. P. (2021). Numerical methods for engineers (8th ed.). McGraw-Hill Education.

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Title	Incompressible Fluid Flow	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	MTech and PhD	Type	C
Prerequisite			

Objectives:

1. To provide a deeper and more thorough grounding in principles and basic applications of fluid mechanics to Mechanical Engineering students.
2. Enhanced understanding of fluid dynamics, including the equations of motion in differential and integral form.
3. Through exposure to the concepts of Boundary layer, turbulence etc.

Learning Outcomes:

1. Ability to predict and solve engineering problems using conservation of mass and momentum.
2. Enhanced analytical ability in solving multi-scale fluid mechanics problems.
3. For an application involving fluid mechanics, be able to utilize a commercial CFD software program in the problem solution.

Contents:

1. Introduction: definition of compressibility, incompressible and compressible flows; Review on vector calculus and tensors **[4 lectures]**
2. Fundamentals: Fluid kinematics; stress-deformation relation, stream function, potential function, vorticity and circulation; Equations of fluid motion: Integral and differential approaches, Euler equation for inviscid flows and Bernoulli's equation, Navier-Stokes equations for viscous flows; Exact solutions: Couette flow, Hagen-Poiseuille flow, flow through circular pipes **[13 lectures]**
3. Vorticity Dynamics: Kelvin's circulation theorem, Vorticity transport equation **[4 lectures]**
4. Irrotational flows: potential theory for elementary flows, application of complex variables in Irrotational flows **[3 lectures]**
5. Boundary layer theory: Von-Karman integral equation, Blassius solution, effect of pressure gradient, separation and control **[4 lectures]**
6. Flow instability: concept of small-perturbations, linearized stability of parallel viscous flows, Orr-Sommerfeld equation **[4 lectures]**
7. Introduction to turbulence: Reynolds averaged Navier-Stokes equation, turbulent kinetic energy and dissipation, fully developed turbulent flow through a pipe and channel, universal law of wall, effect of wall roughness **[4 lectures]**
8. Introduction to compressible flows **[3 lectures]**

Text Books:

1. Kundu P. K., Kohen I. M, and Dowling D. R (2014), Fluid Mechanics, Academic Press.
2. White F. M. (2015), Fluid Mechanics, McGraw Hill.

Reference Books:

1. Fox, R. W., Pitchard, P. J and McDonald, A. T., (2010), Introduction to Fluid Mechanics, 7th Edition, John Wiley & Sons Inc.

Online Material:

1. National Committee for Fluid Mechanics Films: http://web.mit.edu/hml/ncfmf.html

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Title	Applied Heat and Mass Transfer	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	MTech and PhD	Type	C
Prerequisite			

Objectives:

1. To learn the concepts and design methodologies involved in various types of heat and mass transfer applications.
2. To apply the concept of heat and mass transfer to design the system.

Learning Outcomes:

1. Interpret solutions to heat and mass transfer problems.
2. Ability to analyze and select the heat and mass transfer devices.
3. Design the system related to heat and mass transfer application.

Contents:

1. Generalities and fundamentals: Heat transfer by conduction, convection and radiation **[4 lectures]**
2. Heat Conduction: basic law, governing equations in differential form, solution methods, steady state, unsteady state problems-fins, moving boundaries **[6 lectures]**
3. Convective heat transfer: conservation equations, boundary layer approximations, Forced convective laminar and turbulent flow solutions, Natural convection solutions, correlations **[6 lectures]**
4. Boiling and Condensation: Basic definitions, Inception of Boiling in channel Flow, boiling map, Flow models, prediction of friction factor during flow boiling, Prediction of Heat transfer, Nusselt's theory of film condensation, condensation on tubes and tubes banks, In-tube condensation, drop wise condensation **[6 lectures]**
5. Heat Exchanger Design: Design of Tube-in-Tube Parallel and Counter Flow Heat Exchanger, Design of Shell and Tube Heat Exchanger, Introduction to Compact Heat Exchanger, Analysis of Compact heat exchanger **[4 lectures]**
6. Heat Transfer Enhancement: Modes of enhancement, Different methods of Enhancement, Introduction to different passive methods of heat transfer enhancement, Mechanism of heat transfer through surface enhancement, Mechanism of heat transfer through inserts, Introduction to different active methods of heat transfer enhancement **[6 lectures]**
7. Radiation heat transfer: mechanism, properties, exchange between black and nonblack surfaces **[3 lectures]**
8. Mass Transfer: governing laws, Mass transfer coefficients, applications **[4 lectures]**

Text Book:

1. Ganapathy V. (1982), Applied heat transfer, PennWell Publishing.

Reference Books:

2. Eckerst E.R.G. and Drake R.M. (1972), Analysis of Heat Transfer, McGraw-Hill.
3. Faghri A., Zhang Y. and Howell J. (2010), Advanced Heat and Mass Transfer, Global Digital Press.

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Title	Experimental and Computational Techniques	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	MTech and PhD	Type	C
Prerequisite			

Objective:

1. Equip students with a comprehensive understanding of sensing principles, electronics, and data acquisition systems, alongside proficiency in image processing, experimental design, and data analysis to design effective experimental setups and address complex engineering problems.

Learning Outcome:

1. Students will solve and interpret ordinary and partial differential equations, applying numerical methods like Runge-Kutta and finite difference techniques to address real-world engineering problems.

Contents

1. Sensing Fundamentals: Sensing principles, accuracy, resolution, time response, sensitivity and linearity, noise and signal-to-noise ratio, sensor calibration, sensor types and selection **[4 lectures]**
2. Electronics and Data Acquisition: Electronics fundamentals, data acquisition systems, oscilloscopes, signal conditioning, microcontrollers and embedded systems, power supply and grounding, data logging and storage, analog to digital conversion **[4 lectures]**
3. Imaging and Image processing: Cameras, Imaging systems, Image processing, cross-correlation for PIV, Image formats and compression, 3D Imaging, machine vision, illumination techniques **[5 lectures]**
4. Experimental Design: Design of experiments, hypothesis testing, sample size and power analysis, uncertainty quantification, validation and verification, order of magnitude and dimensional analysis, statistical design methods **[4 lectures]**
5. Roots of Equations: Bracketing and open methods, common pitfalls, rate of convergence **[4 lecture]**
6. Linear Algebraic Equations: Sparse and Dense matrices, direct methods - Gauss Elimination and iterative methods - Jacobi, Gauss Seidel, Krylov space methods, rate of convergence, order of complexity **[6 lectures]**
7. Partial Differential Equations: Classification, discretization and stability criteria for PDEs; Consistency, and convergence; Error estimates; Explicit and implicit approaches **[6 lectures]**
8. Data Processing and Analysis: Error analysis, Fast Fourier Transform, Proper Orthogonal Decomposition (POD/DMD)s, Regression and Curve-Fitting **[6 lectures]**

Textbooks:

1. Chapra, S. C., & Canale, R. P. (2021). Numerical methods for engineers (8th ed.). McGraw-Hill Education.
2. Holman, J. P. (2012). Experimental methods for engineers (8th ed.). McGraw-Hill.

Reference books:

1. Eckert, E. R. G., Goldstein, R. J., Pfender, E., & Ibele, W. E. (Eds.). (1970). Measurement techniques in heat transfer (2nd ed.). Technical Press
2. Doebelin, E. O. (2004). Measurement systems: Application and design (5th ed.). McGraw-Hill.
3. Rathakrishnan, E. (2020). Instrumentation, measurements, and experiments in fluids (2nd ed.). CRC Press.

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Title	Thermo-Fluids Lab	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	0-0-2 [1]
Offered for	MTech and PhD	Type	C
Prerequisite			

Objective:

1. Equip students with a comprehensive understanding of sensing principles, electronics, and data acquisition systems, alongside proficiency in image processing, experimental design, and data analysis to design effective experimental setups and address complex engineering problems.

Learning Outcome:

1. Students will solve and interpret ordinary and partial differential equations, applying numerical methods like Runge-Kutta and finite difference techniques to address real-world engineering problems.

Contents:

S. No.	Lab Exercise
1	Sensor principle, calibration, time response, accuracy, sensitivity and SNR
2	Data Acquisition Systems - circuit, data collection, signal conditioning, analog to digital conversion
3	Embedded Systems - use of microcontrollers, embedded programming, embedded storage
4	Principles of cameras - lens selection, illumination techniques, image capturing formats, image filtering
5	Image processing techniques, e.g., cross-correlation algorithms, application for feature detection
6	Design an experiment based on DOE principles, formulation and testing of hypothesis, calculation of sample size, interpret results
7	Order of magnitude or dimensional analysis, uncertainties in measurements, validation
8	Root finding applications - open and closed methods
9	Physical systems to linear algebra problems, solutions of linear systems - direct and iterative methods
10	ODE/PDE representing a physical system, numerical solution based on classification
11	FFT - time domain to frequency domain, error analysis on sensor measurements, propagation of uncertainties, regression for model fitting
12	Singular Value Decomposition - extract modes, Error analysis on decomposition and reconstruction, regression and curve fitting

Textbooks:

1. Chapra, S. C., & Canale, R. P. (2021). Numerical methods for engineers (8th ed.). McGraw-Hill Education.
2. Holman, J. P. (2012). Experimental methods for engineers (8th ed.). McGraw-Hill.

Reference books:

1. Eckert, E. R. G., Goldstein, R. J., Pfender, E., & Ibele, W. E. (Eds.). (1970). Measurement techniques in heat transfer (2nd ed.). Technical Press.
2. Doebelin, E. O. (2004). Measurement systems: Application and design (5th ed.). McGraw-Hill.

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Title	Engineering Thermodynamics	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	M. Tech. and PhD	Type	E
Pre-requisites	Undergraduate thermodynamics, Heat Transfer, Fluid mechanics		

Objectives:

1. Provide students with a comprehensive understanding of advanced thermodynamic principles, including laws, property relations, multiphase systems, and reacting systems, to analyze and optimize engineering processes.
2. Equip students with the ability to apply thermodynamic concepts to practical applications such as energy storage, carbon capture, refrigeration cycles, and fuel cells, emphasizing chemical equilibrium and kinetics.

Learning outcomes:

1. Students will demonstrate proficiency in applying thermodynamic laws, property relations (e.g., Maxwell relations, Clausius-Clapeyron equation), and Gibbs phase rule to analyze entropy, exergy, and equilibrium in multiphase systems and mixtures.
2. Students will be able to evaluate and optimize thermodynamic processes in reacting systems and advanced applications (e.g., adiabatic flame temperature, organic Rankine cycles, supercritical CO_2 systems), using chemical potential, reaction kinetics, and energy minimization principles.

Content:

1. Fundamentals: Review of thermodynamic laws (Zeroth, first, second, third), state properties, processes, ideal gas, thermodynamic cycles, entropy and irreversibility, exergy and available work **[6 lectures]**
2. Thermodynamic property relations: Maxwell relations; Relations involving enthalpy, internal energy and entropy; Mayer relation, Clausius-Clapeyron equation, Joule-Thompson experiment **[6 lectures]**
3. Multiphase systems and mixtures: Energy minimization principle, Gibbs free energy, Helmholtz free energy, Stability, Clapeyron relation, Gibbs phase rule, simple mixtures, thermodynamic relations for mixtures, liquid vapor equilibrium for mixtures, ideal and non-ideal solutions, colligative properties **[9 lectures]**
4. Reacting systems: Chemical equilibrium, chemical potential, Enthalpy of reaction, adiabatic flame temperature, chemical kinetics, Elementary reaction rates, Rates of reaction for multistep mechanisms **[5 lectures]**
5. Statistical Thermodynamics: Introduction, energy states and energy levels, macro and micro-stares, thermodynamic probability, Distribution function, partition energy, statistical interpretation of entropy and temperature **[13 lectures**]

Text Book:

1. Reynolds, W. C., & Colonna, P. (2018). Thermodynamics: Fundamentals and engineering applications. Cambridge University Press.

Reference Books:

1. Bejan, A. (2016). Advanced engineering thermodynamics (4th ed.). Wiley.

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2. Moran, M. J., Shapiro, H. N., Boettner, D. D., & Bailey, M. B. (2018). Fundamentals of engineering thermodynamics (9th ed.). Wiley.
3. Prausnitz, J. M., Lichtenthaler, R. N., & de Azevedo, E. G. (1998). Molecular thermodynamics of fluid-phase equilibria (3rd ed.). Prentice Hall.
4. Smith, J. M., Van Ness, H. C., Abbott, M. M., & Swihart, M. T. (2017). Introduction to chemical engineering thermodynamics (8th ed.). McGraw-Hill Education.

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Title	Convective Heat and Mass Transfer	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B. Tech., M.Tech., and Ph.D.	Type	E
Prerequisite			

Objectives:

1. To provide a deeper and more thorough analytical and numerical background for coupled heat and fluid flow problems relevant to multi-disciplinary engineering applications.
2. Exposure to the concept of thermal and hydrodynamic boundary layers and their interactions.
3. Exposure to solution techniques of solving systems of coupled non-linear PDEs using analytical and numerical methods.

Learning Outcomes:

1. Enhanced analytical and numerical ability to solve a system of coupled non-linear PDEs associated with multidimensional convection problems.
2. Ability to Identify types of boundary conditions and coupling constraints.
3. Ability to produce codes to solve different classes of coupled heat and fluid flow problems.

Contents:

1. Fundamentals: Conservation equations for mass, momentum, energy and species. Boundary layer flows, similarity parameters, dimensionless numbers **[9 lectures]**
2. Laminar Flow Convection: External laminar flow heat transfer. Internal laminar flow heat transfer, entrance region **[9 lectures]**
3. Laminar Flow Convection: Turbulent flow heat transfer, turbulent Prandtl number, external and internal flows **[6 lectures]**
4. Natural Convection: Natural convection in external and bounded flows. Rayleigh-Bénard Convection **[6 lectures]**
5. Mixed Convection: Boiling – pool boiling and forced convection boiling in tubes. Condensation over a plate, tube and tube banks. Mass transfer. Applications to engineering problems **[9 lectures]**

Text Book:

1. Bejan, A. (2013), Convection heat transfer, 4th ed., John Wiley

Reference Books:

1. Kakaç, S., Shah R. K., and Aung, W. (1987), Handbook of single-phase convective heat transfer, John Wiley
2. Kays, W. M. (2005), Convective heat and mass transfer, 4th ed., Tata McGraw-Hill

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Title	Conduction and Radiation	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B. Tech., M.Tech., and Ph.D.	Type	E
Prerequisite			

Objectives:

1. To provide a deeper and more thorough analytical background for 1-D, 2-D, 3-D steady and transient multi-scale heat conduction problems relevant to multi-disciplinary engineering applications.
2. Exposure to the concepts of solving different classes of partial differential equations (PDE) associated with heat conduction problems analytically and numerically.
3. To provide a deeper and more thorough analytical background for radiation heat transfer associated with opaque surfaces and participating media

Learning Outcomes:

1. Enhanced analytical ability to solve PDEs associated with multidimensional conduction problems.
2. Ability to identify types of boundary conditions.
3. Ability to produce codes to solve different classes of conduction problems and familiarity with analytical and empirical radiation models used in CFD tools.

Contents:

Conduction:

1. Recapitulation, 3-D conduction, isotropic, orthotropic and anisotropic solids, Mathematical formulation, analytical solutions, variation of parameters **[8 Lectures]**
2. Integral method, periodic boundary conditions, Duhamel's theorem and Green's function, Stationary and moving heat sources and sinks, Moving boundary problems **[8 Lectures]**
3. Inverse heat conduction problems, Micro-scale heat transfer, Bio-heat transfer, combined-mode heat transfer, integro-differential equations, introduction to Monte-Carlo technique **[8 Lectures]**

Radiation:

4. Recapitulation, Radiative properties of opaque surfaces, Configuration factors **[5 Lectures]**
5. Enclosure analysis for diffuse-gray surfaces and non-diffuse non-gray surfaces, Radiation in absorbing, emitting and scattering media **[5 Lectures]**
6. Engineering treatment of gas radiation in enclosures, Radiation measurements **[5 Lectures]**

Text Book:

1. Ozisik, Necati M. (1993), Heat conduction, 2nd ed., John Wiley & Sons.

Reference Book:

1. Howell, J. R., Menguc M. P., and Robert Siegel. (2015), Thermal radiation heat transfer, 6th ed., CRC Press.

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Title	Turbulent Flows	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P-Th [C]	3-0-0 [3]
Offered for	MTech and PhD	Type	E
Prerequisite			

Objectives:

1. Introduce the basic properties of turbulence
2. Expose the students to theoretical and numerical techniques to describe and quantify the effects of turbulence

Learning Outcomes:

1. Understand the underlying flow physics of turbulent flows and make a trade-off between different applicable models
2. Carry out effectively CFD simulation of an unsteady turbulent flow

Contents:

1. Introduction to turbulence; fundamental equations of turbulent flow; statistical description of turbulence, Reynolds equations **[8 lectures]**
2. Kolmogorov's theory, scales of turbulence, homogeneous or isotropic turbulence, energy transfer, spectral description **[11 lectures]**
3. Turbulent free-shear flows, bounded flows, boundary layers **[6 lectures]**
4. Simulating turbulent flow, Reynolds averaged Navier-Stokes approach, introduction to large eddy simulations and direct numerical simulation **[10 lectures]**
5. Application of AI in turbulence modeling **[4 lectures]**

Text Book:

1. Pope, S. A. (2000), Turbulence, Cambridge

Reference Books:

1. Davidson P. A. (2015), Turbulent Flows, 2nd ed., Oxford
2. Durbin P. A., Reif B. A. (2001), Statistical Theory and Modeling for Turbulent Flows, John Wiley

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Title	Computational Fluid Dynamics and Heat Transfer	Number	MEL7400
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech, M.Tech and PhD	Type	E
Prerequisite			

Objectives:

1. To develop a strong foundation in the governing equations, numerical methods, and physical modeling of fluid flow and heat transfer.
2. To apply CFD techniques using software tools, with an understanding of discretization, convergence, and solution accuracy.
3. To analyze and interpret CFD results for real engineering problems and practical modeling considerations.

Learning Outcomes:

1. Demonstrate understanding of the key theories, approaches, and methodologies underpinning CFD.
2. Apply standard CFD procedures using modern software tools.
3. Utilize CFD analysis to solve and interpret real-world engineering design problems.

Contents:

1. Introduction to CFD: CFD workflow, Applications of CFD, Review of governing equations for heat transfer and fluid flow; Brief introduction of ODE (IVP and BVP) and PDE, Initial and Boundary conditions, classification of PDE, various methods to solve PDE numerically **[6 Lectures]**
2. Finite Difference Method: Taylor series expansion; derivation of finite difference schemes; accuracy and order of discretization; stability and truncation error analysis of finite difference formulations, Application to heat transfer problems, Steady state heat conduction with heat source, Convective- diffusive system – Upwind scheme, Transient heat conduction: explicit, implicit and Crank – Nicolson methods, Handling variable thermo-physical properties, fluid flow problems, Solution of simultaneous equations: direct and iterative methods; Jacobi and various Gauss-Seidel methods, Gauss-elimination, TDMA (Thomas), Gauss-Jordan, other direct and indirect methods **[12 Lectures]**
3. Finite Volume Method: Finite Volume Method formulation, calculation of fluxes at faces of control volume, Source term linearization Discretization for convective terms: upwind, power law, QUICK schemes, staggered grid and solution of incompressible fluid flow, SIMPLE algorithm and extensions to SIMPLER, SIMPLEC and PISO, multigrid techniques **[12 Lectures]**
4. Advanced Topics: Solution convergence, Verification and Validation, Boundary conditions, Grid generation, Turbulent flows; Reynolds-averaged Navier–Stokes (RANS) equations; Boussinesq approximation; RANS models; Introduction to high-fidelity simulations (LES/DNS), Data driven techniques **[9 Lectures]**

Text Book:

1. Versteeg, H.K, and Malalasekera, W. (2007), Computational fluid Dynamics-The finite volume method, Pearson Education.

Reference Books:

1. Anderson, John, D. (1995), Computational Fluid Dynamics: The Basics with Applications, McGraw Hill.
2. Patankar, S. (1980), Numerical Heat Transfer and Fluid Flow, Hemisphere Publishing Corporation.

Online Material:

1. Introduction to Computational Fluid Dynamics and Principles of Conservation by Prof. S. Chakraborty: https://nptel.ac.in/courses/112105045
   Fluid Mechanics 101 Website: https://www.fluidmechanics101.com/

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Title	Compressible Fluid Flows	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech, M.Tech and PhD	Type	E
Prerequisite			

Objectives:

1. To bring an in-depth understanding on the theory of compressible fluid flows, especially supersonic flows.
2. To introduce shock waves in supersonic flows and its reflection, blast waves and various experimental techniques for high speed flow visualization.

Learning Outcomes:

The students are expected to have the ability to:

1. Predict and solve fluid dynamics of engineering applications involving supersonic flows such as, flow over supersonic vehicles, nozzle flows, rocket plumes etc.
2. Predict flow properties across shock waves and to take up research in high speed flows and shock wave dynamics.
3. Conduct various high speed flow measurement experiments.

Contents:

1. Review of thermodynamics; Governing equations for fluid flow **[3 lectures]**
2. Introduction to compressible flows: sonic velocity and Mach number, wave propagation **[3 lectures]**
3. Isentropic flows: static and stagnation condition, isentropic flows with area change, nozzle operation **[4 lectures]**
4. Steady adiabatic compressible flows: normal shock, RankineHugoniot equations, oblique shock; Prandtl-Meyer expansion flows **[5 lectures]**
5. Adiabatic flow with area change: normal shock in c-d nozzles, oblique shocks in over expanded and under expanded nozzle flows, supersonic wind tunnel **[5 lectures]**
6. Shock reflections: reasons for reflection, Regular and Mach reflections **[3 lectures]**
7. Analytical approach for regular and Mach reflection: two shock and three shock theory, shock polar **[4 lectures]**
8. Transition criteria: von-Neumann and detachment criteria **[2 lectures]**
9. Compressible flow with friction: Fanno Flow **[3 lectures]**
10. Compressible flow with heat transfer: Rayleigh flow **[2 lectures]**
11. Unsteady compressible flows: 1-D wave motion: simple and finite waves, Riemann invariants, wave propagation in shock tube, blast waves; Method of characteristics (MOC) **[3 lectures]**
12. Measurements in compressible flows: Schlieren and shadowgraph **[2 lectures]**

Text Book:

1. Zucker R. D. and Biblarz O. (2002), Fundamentals of Gas Dynamics, 2nd ed., Wiley

Reference Books:

1. John J. E. A. and Keith T (2006)., Gas Dynamics, 3rd ed., Prentice Hall

Anderson J. D. (2020), Modern Compressible Flow: With Historical Perspective, 4th ed., McGraw-Hill

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Title	Nanosensors	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	M. Tech and PhD	Type	E
Prerequisite			

Objectives:

The Instructor will:

1. Provide understanding about the importance of nanoscale materials for sensing applications.
2. Teach approaches for fabrication and characterizing of sensors based on nanomaterials.
3. Enhance critical, creative and innovative thinking to tailor nanomaterials for specific applications.

Learning Outcomes:

The students are expected to have the ability to:

1. Design and fabricate nanomaterial based sensors.
2. Undertake interdisciplinary research and exploit nanomaterials for new sensing applications.

Content:

1. Sensor types: Displacement, position and proximity sensors **[3 lectures]**
2. Velocity, motion, force and pressure sensors **[4 lectures]**
3. Components and classification of sensors **[1 lecture]**
4. Parameters for sensor characterization **[1 lecture]**
5. Sensor arrays **[1 lecture]**
6. Nanostructures and Nanoparticles: Nanostructures **[1 lectures]**
7. nanostructure fabrication (top down and bottom up approach) **[3 lectures]**
8. Characterization of nanostructures using different techniques (Atomic Force Microscopy, Scanning Electron Microscopy, Tunneling Electron Microscopy) **[3 lectures]**
9. Nanoparticles (types and shapes) **[1 lecture]**
10. Nanoparticle production, shape control, functionalization and application **[2 lectures]**
11. Nanowires (properties, sensing mechanism, fabrication, and devices) **[3 lectures]**
12. Quantum dots **[1 lecture]**
13. Carbon nanotubes/nanofibres (CNT/CNF) and other Nanosensors: CNT structure, properties, synthesis, functionalization **[3 lectures]**
14. CNT/CNF based sensors (single layer/multi-layer) **[2 lectures]**
15. Nanostructure, nanoparticles, nanowire and nanofibre based metal oxide sensors **[3 lectures]**
16. Flexible sensors and electronic skin **[2 lectures]**
17. Nanosensors in engineering applications: optical, chemical, mechanical, electrochemical, mass sensitive, biosensors, mechanical sensors in liquid solution, thin film sensors **[5 lectures]**

Text Book:

1. Khanna, V. K., (2011), Nanosensors: Physical, Chemical, and Biological, CRC Press.

Reference Books:

1. Atta, N. F., (2014), Nanosensors: Materials and Technologies. IFSA Publishing.
2. Meyyappan, M., (2004), Carbon Nanotubes: Science and Applications, 1st Edition, CRC Press.

Online Material:

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1. Micro and Smart Systems, IISc Bangalore by Prof. K.N. Bhat, Prof. G.K. Anathasuresh, Prof. S. Gopalakrishnan, Dr. K.J. Vinoy: https://nptel.ac.in/courses/112108092
2. Nanotechnology: A Maker's Course by Nan M. Jokerst, Carrie Donley, James Cahoon, Jacob Jones: https://www.coursera.org/learn/nanotechnology/home

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Title	Probabilistic Methods in Engineering Design	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	MTech and PhD	Type	E
Prerequisite			

Objective:

1. To provide a unified probabilistic approach to design aspects in transport phenomena, hydrology, climatic events, properties of materials and dealing with environmental and ecological variables.

Learning Outcomes:

The students are expected to have the ability to:

1. Ability to model and predict phenomena such as development of strength in composite materials, size effect in failure theories, precipitation events, earth quake occurrences & system behaviors etc.
2. To assess the reliability of engineering structures, to assess hazard rates of extreme climatic events, to assess the multi-parameter influences on flow, electro-kinetic parameters.

Contents:

1. Data: Hydrologic Data, Big data, Overview of how data can be helpful in designing and planning urban spaces, social data, experimental data and specific nature of the data related to heat measurement, electricity, dielectric strength, porosity, particle size distribution, and other materials properties, stochastic data and its behavior, line diagram representation of linear and nonlinear data **[7 lectures]**
2. Prior and Post event categorization: Collective and exhaustive events, Total probability theorem, Bayes Theorem, Calculations of Probability Density Function, Cumulative Density Function when experimental data is provided **[6 lectures]**
3. Data Reduction: Descriptive statistics, Skewness, Moments, Kurtosis, Parameter Estimation, Chebyshev Inequality, Law of Large Numbers, Descriptive statistics introduction and training exercises in design of experiments and data compilation **[6 lectures]**
4. Presentation: Hydrographs, Distribution graph, flood frequency studies, normal annual precipitation events, lognormal models in fracture of materials and strength characterization, Gumbel Distribution, Log Pearson Type III Distribution, Extreme events type 1 distribution, Raleigh Distribution (assessing errors in projectile motion), exponential distribution, Erlangs distribution and its variation the gamma distribution, Markov Processes. MINITAB 15/16 introduction and training exercises on transforming data as required by the software **[7 lectures]**
5. Analysis: Auto-correlation, hydrologic time series, spectral density functions, Regression, removing correlations between predictor variables, linear multi-parameter regression, Principal components, factor analysis, multi-variate regression analysis, Boot strapping and Jack Knife Procedures. Exercises to find correlation matrices from available experimental data, finding principal components using multi-variate analysis etc. Introduction to GEPHI and social network data analysis with intra-levels **[9 lectures]**
6. Testing: Sensitivity Analysis, Kolmogorov Smirnov Tests, Chi square Tests, Analysis of variance. Exercises to perform normality tests and variance analysis. **[4 lectures]**

Text Book:

1. Benjamin J. R. and Cornell A. C. (1970), Probability, Statistics and Decision for Civil Engineers, McGraw Hill

Reference Books:

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1. Ang A. H. S. and Tang W. H. (1975), Probability Concepts in Engineering, Planning and Design Vol1- Engineering and Design, John Wiley
2. Ang A. H. S. and Tang W. H. (1984), Probability Concepts in Engineering, Planning and Design Vol2- Decision, Risk and Reliability, John Wiley

Online Material:

1. Probability Methods in Civil Engineering, IIT Kharagpur by Dr. Rajib Maity: https://nptel.ac.in/courses/105105045/

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Title	Combustion Technology	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	BTech, MTech and PhD	Type	E
Prerequisite			

Objectives:

1. To introduce energy and combustion fundamentals for analyzing modern combustion systems.
2. To develop understanding of the basic principles and concepts of advanced fuel combustion and control process.
3. To be familiar with the fundamental physical and chemical principles regarding formation and control of air pollutants in industrial and technological processes.

Learning Outcomes:

1. Understanding of thermodynamics and kinetics of combustion.
2. Analysis of reacting systems and design of clean combustion systems

Contents:

1. Importance of combustion, Combustion phenomena in gas turbines, IC engines and power plants; Review of thermodynamics, Chemical kinetics, Mass transfer definitions: Fick's law. Equations of conservation of species mass, momentum, and energy; multicomponent diffusion equation, Schvab-Zel'dovich formulation, Rankine-Hugoniot relations **[13 lectures]**
2. Laminar premixed flames: Flame speed, adiabatic flame temperature, flammability limits, flame stabilization, ignition and quenching; Laminar diffusion flames: BurkeSchumann problem and droplet burning; Partially premixed flames **[13 lectures]**
3. Introduction to turbulent flames: premixed and diffusion flames; Detonations and Combustion instabilities **[6 lectures]**
4. Combustion of solids: solid propellant, drying, devolatilization and char combustion Spray combustion **[4 lectures]**
5. Chemical Emission from combustion, Quantification of emission, Emission control methods, Introduction to combustion modeling **[3 lectures]**

Text Book:

1. Turns, S. R. (2021), An Introduction to Combustion: Concepts and Applications, 4th Ed., McGraw-Hill

Reference Books:

1. Warnatz, J., Maas, U., & Dibble, R. W. (2006). Combustion: Physical and chemical fundamentals, modeling and simulation, experiments, pollutant formation (4th ed.). Springer.
2. Kuo K.K. (1986), Principles of Combustion, John Wiley and Sons

Online Material:

1. Fundamentals of Combustion, IIT Madras, Prof. V. Raghavan : https://nptel.ac.in/courses/112106299

भारतीय प्रौद्योगिकी संस्थान जोधपुर
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Indian Institute of Technology Jodhpur
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Title	Design of Heat Exchange Equipment	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	MTech and PhD	Type	E
Prerequisite			

Objectives:

1. To learn the concepts and design methodologies involved in various types of Heat exchange devices.
2. To gain a deeper understanding and design process associated with heat exchangers.

Learning Outcomes:

1. Learn methods of heat transfer in a heat exchange system.
2. Develop ability to design and performance assessment of heat exchanger devices.
3. Develop codes for designing heat transfer systems.

Contents:

Construction Details and Heat Transfer Analysis:

1. Types of Heat exchange equipment used in industries, General design consideration of heat exchangers, Double pipe heat exchanger, Shell and Tube Heat Exchangers, Regenerators and Recuperators, Industrial applications **[7 Lectures]**
2. Temperature distribution and its implications, LMTD, Effectiveness **[4 Lectures]**
3. Flow Distribution and Stress Analysis, Effect of Turbulence, Friction factor, Pressure loss, Channel divergence, Thermal Stress in tubes, Types of failures **[6 Lectures]**

Design Aspects:

1. Calculation of Heat Transfer and pressure loss, Flow Configuration, Effect of Baffles, effect of Deviations from ideality, Design of Typical liquid, Gas-Gas-Liquid Heat Exchangers, Plate Heat Exchangers **[7 Lectures]**
2. Condensers and Evaporators Design: Design of Surface and Evaporative Condensers, Design of Shell and Tube, Plate type evaporators **[5 Lectures]**
3. Design aspect of Horizontal and Vertical Heat Exchange systems with phase change **[5 Lectures]**
4. Cooling Towers Packings, Spray design, Selection of pumps, Fans and Pipes, Testing and Maintenance, Experimental Methods **[5 Lectures]**

Text Book:

1. Hewuttm G. F., Shires G. L. and Bott T. R. (1994), Process Heat Transfer, CRC Press

Reference Books:

1. Shah R. K., and Sekulic D. P. (2003), Fundamentals of Heat Exchanger Design, John Wiley
2. Smith E. M (2005), Advances in Thermal Design of Heat Exchangers, John Wiley

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Title	Thermal Environmental Engineering	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	MTech and PhD	Type	E
Prerequisite			

Objectives:

1. Develop a comprehensive understanding of refrigeration, heat pump, and air-conditioning technologies, including conventional and advanced heat-driven R&AC systems
2. To understand principles, design methodologies, and performance evaluation of vapour compression, absorption, and hybrid refrigeration and heat pump systems
3. To understand natural refrigerant-based cooling and heating technologies and applications to sustainable thermal systems
4. Enable students to design and evaluate refrigeration and heat pump systems for industrial, commercial, and residential applications using case studies, numerical modelling, and practical exposure

Learning Outcomes:

1. Analyse and evaluate the performance of vapour compression refrigeration systems
2. Model and assess absorption refrigeration and heat pump systems, including advanced cycle configurations
3. Apply psychrometric principles to analyse R&AC processes and components
4. Perform system design calculations, component selection, and performance analysis of green refrigeration, air-conditioning, and heat pump applications

Contents:

1. Introduction: Need for refrigeration, thermal processes, various refrigeration cycles, industrial thermal processes for cooling and heating, case studies **[3 Lectures]**
2. Vapour compression refrigeration and heat pump systems: Vapour compression refrigeration cycle, signe-stage, two-stage, cascade cycles, real systems with pressure drop and heat interactions **[4 Lectures]**
3. Absorption refrigeration and heat pump systems: LiBr/H2O systems with single effect, double effect series and parallel circuits, NH3/H2O single effect, GAX and branched GAX, absorption-resorption and hybrid cycles, heat pump, case studies **[10 Lectures]**
4. Advanced Psychrometric: Psychrometric chart and processes, properties of moist air, air conditioning processes and energy calculations, direct and indirect evaporative cooling, desiccant processes **[5 Lectures]**
5. Natural refrigerant-based refrigeration and heat pump systems: R744/R290 based cooling and heating systems, system design, component selections, performance evaluation, case studies **[8 Lectures]**
6. Introduction to Advanced Refrigeration Cycles: Multi-utility heat pumps, cogeneration of power and refrigeration, solar refrigeration and air-conditioning **[4 Lectures]**
7. Special topics/projects: Guest lectures, site visits, case studies, open-source numerical modelling

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and system designing **[5 Lectures]**

Text Book:

1. Herold K.E., Radermacher R., Klein S.A. (2016), Absorption Chillers and Heat Pumps, CRC Press

Reference Books:

1. 2021 or 2025 ASHRAE Handbook - Fundamentals
2. 2020 or 2025 ASHRAE Handbook - HVAC Systems and Equipment

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Title	Machine Learning for Thermofluids	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	MTech and PhD	Type	E
Prerequisite			

Objectives:

1. To develop a deep understanding of how machine learning (ML) techniques can be systematically integrated with thermofluid physics to enhance modeling, prediction, and design of thermofluidic systems.
2. To prepare students to critically evaluate and apply state-of-the-art machine learning tools for thermofluid research and industrial applications.

Learning Outcomes:

1. Understand core ML algorithms and how to adapt them to thermofluid problems.
2. Evaluate the quality, structure, and relevance of data for ML in thermofluid applications.
3. Develop interpretable, physically-consistent ML models grounded in reliable experimental or simulation data.

Contents:

1. Core Concepts in Machine Learning for Thermofluids: Data Preprocessing, Feature Engineering, Linear Regression for Drag and Heat Transfer Prediction, PCA for Flow and Thermal Field Dimensionality Reduction, Neural Network, Convolutional Neural Networks, Loss Functions, Evaluation Metrics. **[5 Lectures]**
2. Data Generation and Modeling in Thermofluids: Navier-Stokes (NS) Equations, Energy Equation for Heat Transfer, Computational Simulations, Experimental Techniques, Reynolds Averaging for Turbulent Flows, Canonical Flows and Datasets, ML Design pipeline for optimization. **[5 Lectures]**
3. Reduced Order Models: Proper Orthogonal Decomposition, Dynamic Mode Decomposition, Application: Vortex Shedding and Heat Transfer Analysis. **[4 Lectures]**
4. Neural Network Architectures: Physics Informed Neural Networks (PINNs) for Solving NS Equations, Custom Loss Functions with Physical Constraints, Recurrent Neural Networks (RNNs) and LSTMs for Unsteady Flow Dynamics, Autoencoders and GANs for Flow and Temperature Field Reconstruction, Transformers for Long-Term Turbulent Flow Prediction. **[9 Lectures]**
5. Experimental Data and ML Integration: Particle Image Velocimetry (PIV) Data Processing for Turbulent Boundary Layers, Digital Twin, Sensor Data Interpretation for Real-Time Flow and Temperature Monitoring, Applications to Wind Tunnel and Hydrodynamic Testing. **[4 Lectures]**
6. Turbulence Closure Models: Explicit Algebraic Reynolds Stress Models, Tensor Basis Neural Networks, Gene Expression Programming (GEP), Frozen and Computational Fluid Dynamics-Driven GEP for Canonical Flows **[6 Lectures]**
7. Case Studies: Wake Modelling, Airfoils, Heat Exchanger Design, Gas Turbines, Renewable Energy Systems, Flow Control, Additional case studies based on latest developments and not restricted to the above cases, Challenges and future opportunities. **[6 lectures]**

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Text Books:

1. Géron, A. (2022). Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow. O'Reilly Media, Inc.
2. Versteeg, H. K. (2007). An introduction to computational fluid dynamics the finite volume method, 2/E. Pearson.
3. Holman J.P., (2020). Heat Transfer. McGraw Hill.

Reference Books:

1. Mendez, M. A., Ianiro, A., Noack, B. R., & Brunton, S. L. (Eds.). (2023). Data-driven fluid mechanics: combining first principles and machine learning. Cambridge University Press.
2. Brunton, S. L., & Kutz, J. N. (2022). Data-driven science and engineering: Machine learning, dynamical systems, and control. Cambridge University Press

Online Material:

1. Videos by Steve Brunton: https://www.youtube.com/@Eigensteve